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SZZX-ID Bridge Program Physics

Unit 2 Lesson 5: Newton' s Laws of motion

Class Outline

Title of Unit	Forces and Newton's Laws
Title of Lesson	Newton's laws
Starter Activity	Newton as a polymaths
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- inquiry: Newton as a polymath - inquiry and question: sample problems - Collaboration: Kahoot on Newton's third law
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	Group work and discussion on the application of Newton's third law – car design
	International Mindedness:
	Newton as a polymath – boundary between science subjects are arbitrary
Lesson Objectives	By the end of the lesson, students will be able to: • Understand the meaning and mathematical expression of Newton's laws • Understand the definition of force • Able to apply Newton's laws in free body diagram and solve the motion status of an object from the free-body diagram
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Communication: clearly communicate complex ideas
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Thinker, inquiry
Teaching Activities	Group Discussion: design of a sail cart
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Summarize the three laws of motion

Vocabulary



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Exert 施加（力）

Inertia 惯性

Frame of reference 参考系

Action force 作用力

Reaction force 反作用力

Interaction 相互作用

Push 推

Pull 拉

Resist 抵抗

Proportional 成正比的

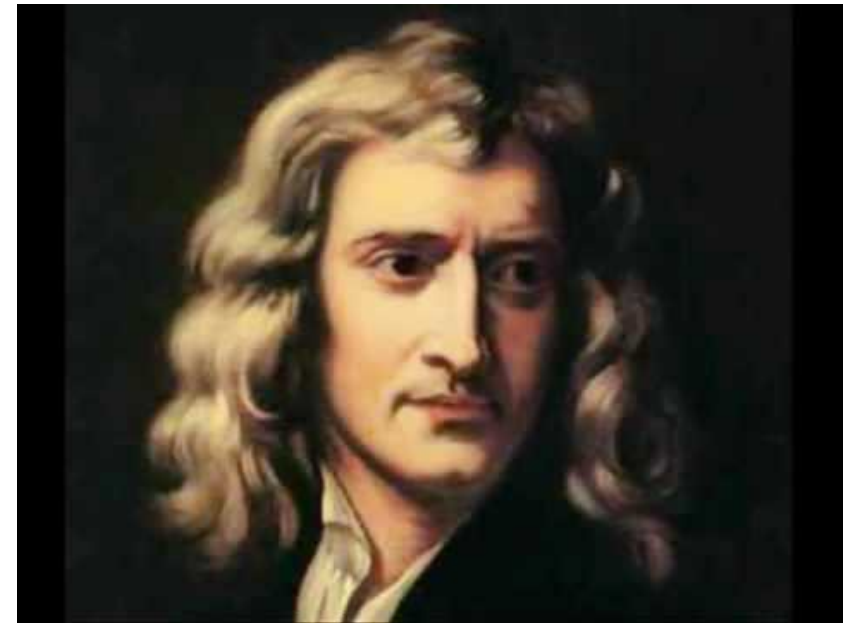
Accelerate 加速

Background



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- Sir Isaac Newton, an English scientist and mathematician famous for his discovery of the law of gravity also discovered the three *laws of motion*. Today these laws are known as *Newton's Laws of Motion* and describe the motion of all objects on the scale we experience in our everyday lives.
- Newton's first Law – the definition of force
- Newton's second Law – the effect of force (acceleration)
- Newton's third Law – the nature of force (action and reaction)



Newton's Third Law



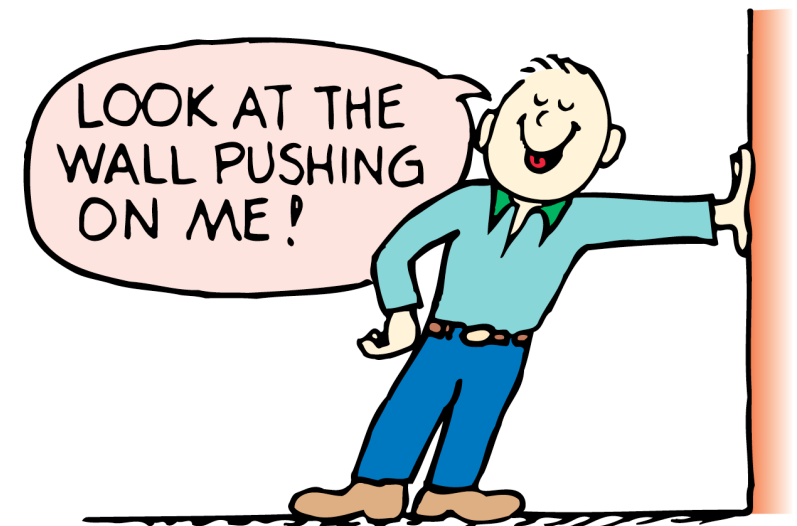
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For every force (action), there is always an equal, and opposite force (reaction)

- Forces result from interactions
- All forces come in pairs
- The action and reaction are acted on different bodies
- The two forces will appear and disappear at the same time
- The action and reaction forces are of same magnitude and opposite direction, along the same line

Mathematically, if we write the force of object 1 on object 2 as F_{12} and object 2 on object 1 as F_{21} , then

$$F_{12} = -F_{21}$$

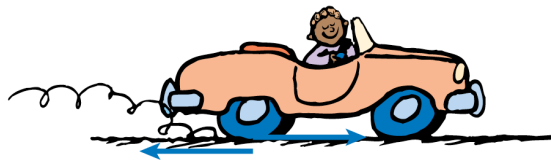


Newton's Third Law

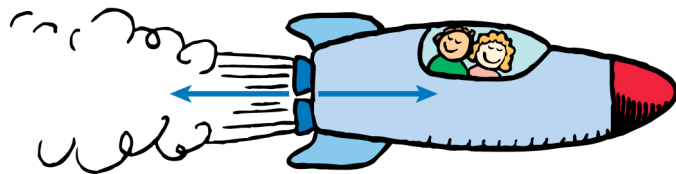


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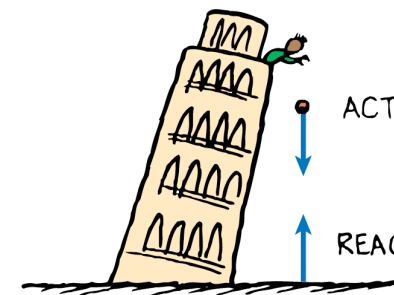
When action is *A exerts force on B*, the reaction is simply *B exerts force on A*.



ACTION: TIRE PUSHES ROAD REACTION: ROAD PUSHES TIRE



ACTION: ROCKET PUSHES GAS REACTION: GAS PUSHES ROCKET



ACTION: EARTH PULLS BALL

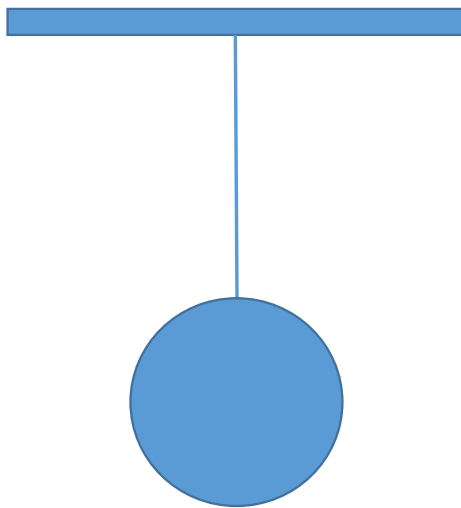
REACTION: BALL PULLS EARTH

Newton's Third law - examples



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Which of the following is a pair of action-reaction forces?

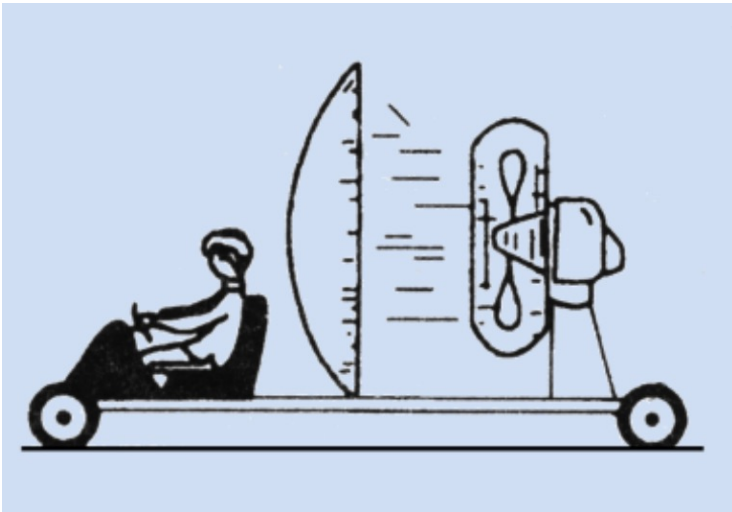


- A. The pulling force exerted by the string to the ball, and the force of gravity of the ball
- ☒ B. The pulling force exerted by the ball to the string, and the pulling force exerted by the string to the ball
- C. The pulling force exerted by the string to the ceiling, and the pulling force exerted by the ball to the string
- D. The pulling force exerted by the string to the ceiling, and the force of gravity of the ball

Newton's Third Law – discussion



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Someone designed a car like this.

Explain why or why not the car can move, using Newton's third law of motion.

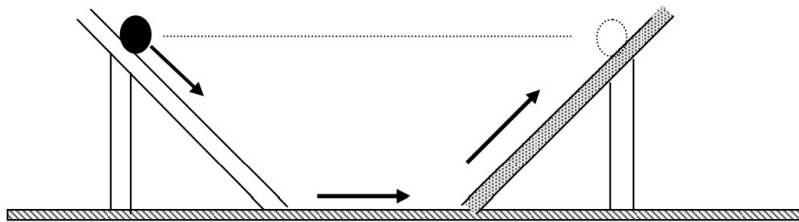
Newton's First Law (Law of Inertia)



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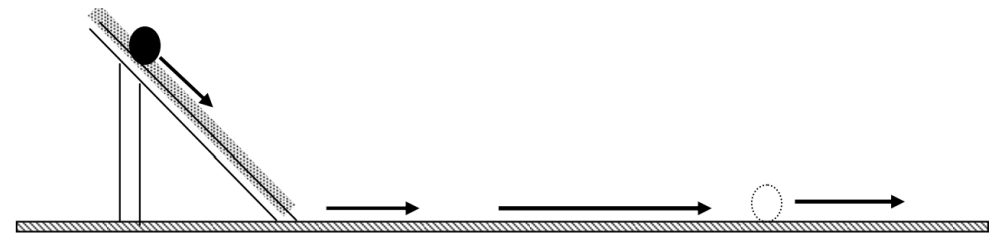
Galilei's work and thoughts

In the early 1600's, the Italian Physicist Galileo Galilei made one of the most important discoveries in history. Galileo discovered that when he rolled a ball down a smooth wooden ramp that the ball would continue rolling until it rolled up a second wooden ramp to the ball's initial height regardless of the second ramp's angle of inclination.



Galileo's Inclined Plane Experiment

Galileo further realized that the floor could be considered as a ramp with a zero angle of inclination (flat). Since the ball would never reach its original height, the ball would continue rolling forever!!



Ball Never Stops!!

Newton's First Law (Law of Inertia)



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A particle will continue to move in a straight line at constant speed unless the particle is acted upon by a net external force.

- Straight line at constant speed = constant velocity
 - Force is not the reason for motion. Force is the reason for the change of motion
 - Newton's first law defines what a force is. A force is what changes a body's velocity (or gives the object a non-zero acceleration)
- ◆ Basically, an object will “keep doing what it was doing” unless acted on by an unbalanced force.
- ◆ If the object was sitting still, it will remain stationary. If it was moving at a constant velocity, it will keep moving.
- ◆ It takes force to change the motion of an object.



Newton's First Law (Law of Inertia)



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- If objects in motion tend to stay in motion, why don't moving objects keep moving forever?

A book sliding across a table slows down and stops because of the force of *friction*.



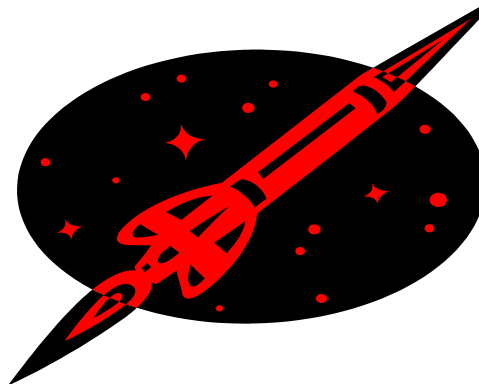
If you throw a ball upwards it will eventually slow down and fall because of the force of *gravity*.

Newton's First Law (Law of Inertia)



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In outer space, away from gravity and any sources of friction, a rocket ship launched with a certain speed and direction would keep going in that same direction and at that same speed forever.



Newton's First Law (Law of Inertia)



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Question: Is inertia a force?

Question: What is the relation between inertia and mass?

Newton's Second Law



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The net force on a body of constant mass is proportional to that body's acceleration and is in the same direction as the acceleration.

Mathematically,

$$\mathbf{F} = m\mathbf{a}$$

- F in the equation is net force. Acceleration will take the same direction of net force.
- For a given object, acceleration is proportional to net force applied
- If we apply the same net force to objects, the accelerations of the objects will be inversely proportional to their mass
- An unbalanced force causes an object to accelerate
- ◆ Unit of force is defined by Newton's second law, when we take kg as the unit of mass and $\text{m} \cdot \text{s}^{-2}$ as unit of acceleration.
- ◆ 1 newton (N) is the force required to accelerate a mass of 1kg by $1 \text{ m} \cdot \text{s}^{-2}$ in the direction of the force

Newton's Second Law



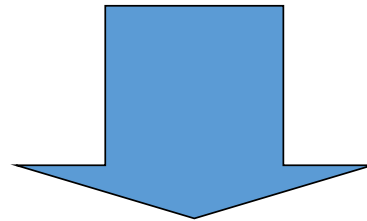
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$$\begin{array}{c} \rightarrow + \leftarrow = \rightarrow \\ \text{Net Force} = \rightarrow \end{array}$$



$$\begin{array}{c} \rightarrow + \rightarrow = \rightarrow \\ \text{Net Force} = \rightarrow \end{array}$$



Unbalanced forces cause acceleration.

Let's look at mass and weight again



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- **Mass** is the quantity of matter in an object. More specifically, mass is a measure of the inertia, or “**laziness**,” that an object exhibits in response to any effort made to start it, stop it, or otherwise change its state of motion.
- **Weight** is the force of gravity on an object
- If force is equal to mass times acceleration, Weight is equal to mass times acceleration due to gravity

$$\begin{array}{ccccc} F & = & m & a \\ \downarrow & & \downarrow & \downarrow \\ W & = & m & g \end{array}$$

Newton's Second Law – exercise 1



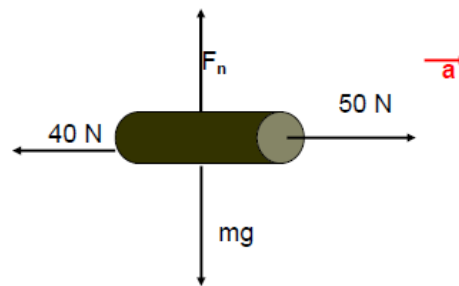
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A 50 N applied force drags an 8.16 kg log to the right across a horizontal surface. What is the acceleration of the log if the force of friction is 40.0 N?

Exam tip

To solve an ' $F=ma$ ' problem:

- Make a diagram.
- Identify the forces on the body of interest.
- Find the net force on each body, taking the direction of acceleration to be the positive direction.
- Apply $F_{\text{net}} = ma$ to each body.



$$F_{\text{net}} = ma$$

Let F_a be the applied force and f be the force of friction.

$$F_a - f = ma$$

$$50 - 40 = 8.16a$$

$$a = 1.23 \text{ m} \cdot \text{s}^{-2}$$

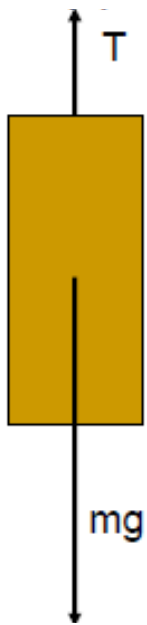
With the same direction of the applied force.

Newton's Second Law – exercise 2

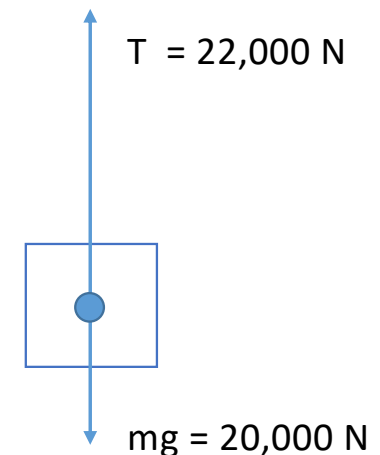


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An elevator with a mass of 2,000 kg rises with an acceleration of 1.00 ms^{-2} .
What is the tension in the supporting cable? Use $g = 10 \text{ ms}^{-2}$



$$\begin{aligned} F_{net} &= ma \\ T - mg &= ma \\ T &= ma + mg \\ T &= 2000 \cdot 1 + 2000 \cdot 10 \\ T &= 22,000 \text{ N} \end{aligned}$$



$$T > mg$$

Newton's Second Law – discussion



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Question:

We know from Newton's second law that force is the reason for acceleration. However, when I try to push a car, I applied force to the car but the car doesn't move. Is it a contradiction of the Second Law?

Kinematics and Newton' s Laws – exercise



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A car with a mass of 1000kg is traveling at a constant velocity of 100 km/h. Then the engine is stopped. The car slides for 60 s before coming to a complete stop.

What is the force of friction when the car is sliding?

When the engine restarts, it supplies a force of 2000N, what will be the acceleration of the car?

Kinematics and Newton's Laws – exercise



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1) Initial velocity of the car $u = 100\text{km/h} = 27.8\text{m/s}$

$$a = (v-u)/t = -27.8/60 = -0.463 \text{ m/s}^2$$

When the car slides, the net force is dynamic friction force f .

$$f = ma = 1000 * (-0.463) = -463 \text{ N}$$

Force of friction is 463 N, pointing to the opposite direction of the initial velocity

2) Net force $F_{\text{net}} = 2000 - 463 = 1537 \text{ N}$

$$F_{\text{net}} = m a$$

$$a = F_{\text{net}} / m = 1537/1000 = 1.537 \text{ m/s}^2, \text{ same direction with the initial velocity}$$

